CoE model vs Migkas+2020 Fig. 6 $A(\ell)$: term matching at fixed $D/r = 2, 3, 4$

Sedov–Taylor Center-of-Expansion model, sources on the vision shell ($d = r$). Analytic point-mass form and numeric ideal-drag model, each with Chris's rotation and 2ℓ transforms. Scripts: `figures/termmatch_Dr234.py` (solve) and `make_termmatch_tex.py` (writes this sheet); every number here is read from `termmatch_Dr234_results.json`.

Step 0. Chris's variables

r = vision-shell radius = 14.3 Gpc (sources at $d = r$); D = our distance from the CoE; L = density scale length; $M = M(< D)$ = mass enclosed toward the CoE; (ℓ_0, b_0) = Galactic longitude and tilt of the CoE axis; φ = rotation phase in $\mathrm{Amp}_1 \sin(\ell + \varphi)$; $\varphi_2 = 2\ell$ phase in $\mathrm{Amp}_2 \sin(2\ell + \varphi_2)$.

Step 1. Define parameters

$$\varepsilon \equiv r/D = 1/(D/r) \;\; (= \tfrac{1}{2},\ \tfrac{1}{3},\ \tfrac{1}{4} \text{ here}), \qquad x \equiv D/L, \qquad M_{\rm crit}(< D) = \tfrac{4\pi}{3}\rho_c D^3 = \frac{H_0^2 D^3}{2G},$$

$$\kappa \equiv \frac{G\,M(< D)}{H_0^2 D^3} = \frac{M(< D)}{2M_{\rm crit}}, \qquad K \equiv \alpha\kappa \;\; (\alpha = O(1) \text{ time-integration factor; tables use } \alpha = 1),$$

$\phi_1 \equiv$ rotation: shift of the m=1 peak, $\qquad \phi_2 \equiv$ shift of the m=2 peak (the 2ℓ transform),

$$\varphi = 90^\circ - (\ell_{\rm nat}^{(1)} + \phi_1), \qquad \varphi_2 = 90^\circ - 2(\ell_{\rm nat}^{(2)} + \phi_2),$$

$$\delta_3 \equiv \ell_3^{\rm data} - \ell_3^{\rm model} \;\; \text{(octupole left at its natural phase; } \ell_3 \text{ defined mod } 120^\circ).$$

Analytic model (point-mass tidal series to $O(\varepsilon)$, averaged over 90° sectors every 5°, $|b| > 20^\circ$ cut, $\cos b$ weight; same as the v2 sheet), with $\psi = \ell - \ell_0$:

$$\frac{A}{A_{\rm all}} = N\big[1 + K(3\mu^2 - 1) + K\varepsilon(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3)\big], \qquad \mu = \cos b \cos b_0 \cos\psi + \sin b \sin b_0$$

$$\xrightarrow{\text{sector avg}} \; N\big[1 + Kq_0\big] + NK\varepsilon\, o_1 \cos\psi + NKq_2 \cos 2\psi + NK\varepsilon\, o_3 \cos 3\psi,$$

$q_0 = 0.45900 - 0.68850a^2,\ q_2 = 0.49052a^2,\ o_1 = 0.90032\,a(-3.1958 + 3.9948a^2),\ o_3 = -0.22806a^3,\ a = \cos b_0.$
Numeric model: ideal-drag ST background, $du/dt + Hu = -GM(< r)/r^2$ integrated in time, density $\rho \propto (1 - f_u)e^{-R/L}/e^{-D/L} + f_u$ (exp core + floor), $H_{\rm obs}(\hat n) = H + [(\vec u(\vec r) - \vec u(D)) \cdot \hat n]/d$, 90° boxcar, mean-normalised; parameters $\kappa,\ D/L,\ f_u,\ b_0,\ \ell_0$ (as in `r2_vs_Dr_density_families.py`; $|b_0| \le 30^\circ$, $D/L \le 300$).

Step 2. Data harmonics (Fourier fit $A = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$, $R^2 = 0.915$)

$$|A_0| = 0.99650, \quad |A_1| = 0.05253\ (\ell_1 = 55.70^\circ), \quad |A_2| = 0.07338\ (\ell_2 = 14.96^\circ), \quad |A_3| = 0.03200\ (\ell_3 = 66.96^\circ)$$

Weights: 1σ amplitude errors from the 71-point least-squares Fourier fit with per-point errors $(\sigma_{\rm lo} + \sigma_{\rm hi})/2$ (overlapping sectors treated as independent, so these are too small by $\sim \sqrt{71/16}$; only their *ratios* matter for the weights): $\sigma_0 = 0.0115,\ \sigma_1 = 0.0156,\ \sigma_2 = 0.0160,\ \sigma_3 = 0.0157$. Axis: $\ell_0 \equiv \ell_2 = 14.96^\circ$.

Step 3. Method

A (analytic, fixed ε). Unknowns N, K, b_0; four amplitude equations, so weighted least squares. Write $P \equiv NK$. The m=0 equation contains N alone and is satisfied exactly for any (P, b_0) by $N = A_0 - Pq_0$. For fixed b_0 the m=1, 2, 3 equations are linear in P: with $c_1 = \varepsilon|o_1|,\ c_2 = |q_2|,\ c_3 = \varepsilon|o_3|$ and $w_m = 1/\sigma_m^2$,

$$P(b_0) = \frac{\sum_m w_m A_m c_m}{\sum_m w_m c_m^2}, \qquad \chi^2(b_0) = \sum_{m=1}^{3} w_m \big(A_m - P\,c_m\big)^2 \;\; \text{minimised over } b_0 \in [0^\circ, 90^\circ)$$

($\pm b_0$ give identical curves, so only $b_0 \ge 0$ is scanned). Transforms: $\phi_1 = \ell_1^{\rm data} - \ell_{\rm nat}^{(1)}$, where $\ell_{\rm nat}^{(1)} = \ell_0$ if $o_1 > 0$ and $\ell_0 + 180^\circ$ if $o_1 < 0$; $\phi_2 = \ell_2 - \ell_0 = 0$ by construction. Octupole: $o_3 < 0$, so its natural peak is $\ell_0 + 60^\circ = 74.96^\circ$, giving $\delta_3 = -8.00^\circ$ at every D/r.

B (numeric). Numeric curve on a 181-point grid, FFT → harmonics $m = 0..3$ about their natural peaks; the m=1 and m=2 components are shifted by ϕ_1, ϕ_2, the rest (octupole) untouched. $(\kappa, D/L, f_u, b_0, \phi_1, \phi_2)$ are fitted to maximise R^2 of the transformed curve vs the 71 points, with ℓ_0 held at 14.96° as in A (a free ℓ_0 only moves the octupole, i.e. it would act as a third free phase; that variant is listed at the end for reference). Start points: the stored no-transform best fits, one differential-evolution run and 4 random starts, all polished with Nelder–Mead.

$D/r = 2$ ($\varepsilon = 0.5000$, $D = 28.6$ Gpc $= 93.3$ Gly)

A. Analytic: set equal (numbers substituted, $\varepsilon = 0.5000$)

$$0.99650 = N(1 + Kq_0), \quad 0.05253 = NK \cdot 0.5000\,|o_1|, \quad 0.07338 = NK\,|q_2|, \quad 0.03200 = NK \cdot 0.5000\,|o_3|$$

Solve. $\chi^2(b_0)$ has 2 local minima on $[0°, 90°)$:

branch	b_0	q_0	q_2	o_1	o_3	$c_1,\ c_2,\ c_3$	$P = NK$	χ^2	dipole
primary	6.26°	−0.2213	0.4847	+0.6725	−0.2240	0.3362, 0.4847, 0.1120	0.15772	0.872	on axis
second	36.18°	+0.0105	0.3195	−0.4312	−0.1199	0.2156, 0.3195, 0.0600	0.24136	1.304	flipped

Primary ($b_0 = 6.26°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{226.668}{1437.15} = 0.15772$, $N = 0.99650 + 0.15772 \times 0.22130 = 1.03140$, $K = P/N = 0.15292$.
Plug back: $m0$: $1.03140(1 - 0.03384) = 0.99650$; $m1$: $0.15772 \times 0.33623 = 0.05303$; $m2$: $0.15772 \times 0.48468 = 0.07644$; $m3$: $0.15772 \times 0.11200 = 0.01766$.
Residuals (model−data): $m0$: +0.00000, $m1$: +0.00050 (+0.03σ), $m2$: +0.00306 (+0.19σ), $m3$: −0.01433 (−0.91σ).
Phases: natural dipole peak $14.96° \Rightarrow \phi_1 = 55.70 - 14.96 = +40.75°$ (+0.711 rad), Chris's φ: $75.04° \to 34.30°$; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.884**; no transforms: 0.752; (octupole also given its own phase: 0.894). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.306/\alpha$.

Second branch ($b_0 = 36.18°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{146.331}{606.27} = 0.24136$, $N = 0.99650 - 0.24136 \times 0.01047 = 0.99397$, $K = P/N = 0.24283$.
Plug back: $m0$: $0.99397(1 + 0.00254) = 0.99650$; $m1$: $0.24136 \times 0.21561 = 0.05204$; $m2$: $0.24136 \times 0.31955 = 0.07713$; $m3$: $0.24136 \times 0.05996 = 0.01447$.
Residuals (model−data): $m0$: +0.00000, $m1$: −0.00049 (−0.03σ), $m2$: +0.00375 (+0.23σ), $m3$: −0.01753 (−1.12σ).
Phases: natural dipole peak $194.96° \Rightarrow \phi_1 = 55.70 - 194.96 = -139.25°$ (−2.430 rad), Chris's φ: $-104.96° \to 34.30°$; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.876**; no transforms: −0.088; (octupole also given its own phase: 0.883). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.486/\alpha$.

B. Numeric: fitted parameters $\kappa = 0.1081$, $D/L = 300$ (bound: point-mass limit), $f_u = 0.0345$, $b_0 = 0.00°$, $\ell_0 = 14.96°$ (fixed), $\phi_1 = -139.78°$, $\phi_2 = -0.12°$.

Set equal (harmonics of the fitted numeric curve vs data)

m	data $\lvert A_m\rvert$	numeric $\lvert M_m\rvert$	resid.	resid./σ_m	natural peak	transformed peak (data)
0	0.9965	1.0000	+0.0035	+0.30	–	–
1	0.0525	0.0462	−0.0063	−0.40	194.96°	55.18° (55.70°)
2	0.0734	0.0798	+0.0064	+0.40	14.96°	14.84° (14.96°)
3	0.0320	0.0207	−0.0113	−0.72	74.96°	74.96° (66.96°)†

†octupole untouched. (The numeric curve is mean-normalised, so its m=0 is 1 by construction; R^2 uses the curve as is, no rescaling.) Octupole mismatch $\delta_3 = -8.00°$. Chris's φ: $-104.96° \to 34.82°$ ($\phi_1 = -2.440$ rad); $\varphi_2 = 60.33°$.
R^2 (71 pts): transformed **0.885**; same parameters, no transforms 0.027; stored no-transform best fit 0.765. $M(< D)/M_{\rm crit} = 2\kappa = 0.216$.
Best solution on the other dipole branch (natural dipole on the axis): $R^2 = 0.879$ (no transforms 0.750), $\kappa = 0.264$, $D/L = 15.24$, $f_u = 0.679$, $b_0 = 30.0°$, $\phi_1 = +40.1°$, $\phi_2 = -0.15°$.

$D/r = 3$ ($\varepsilon = 0.3333$, $D = 42.9$ Gpc $= 139.9$ Gly)

A. Analytic: set equal (numbers substituted, $\varepsilon = 0.3333$)

$$0.99650 = N(1 + Kq_0), \quad 0.05253 = NK \cdot 0.3333\,|o_1|, \quad 0.07338 = NK\,|q_2|, \quad 0.03200 = NK \cdot 0.3333\,|o_3|$$

Solve. $\chi^2(b_0)$ has 2 local minima on $[0°, 90°)$:

branch	b_0	q_0	q_2	o_1	o_3	$c_1,\ c_2,\ c_3$	$P = NK$	χ^2	dipole
primary	39.96°	+0.0545	0.2882	−0.5858	−0.1027	0.1953, 0.2882, 0.0342	0.26589	2.172	flipped
second	0.00°	−0.2295	0.4905	+0.7193	−0.2281	0.2398, 0.4905, 0.0760	0.16849	2.437	on axis

Primary ($b_0 = 39.96°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{129.567}{487.29} = 0.26589$, $N = 0.99650 - 0.26589 \times 0.05452 = 0.98200$, $K = P/N = 0.27076$.
Plug back: $m0$: $0.98200(1 + 0.01476) = 0.99650$; $m1$: $0.26589 \times 0.19528 = 0.05192$; $m2$: $0.26589 \times 0.28817 = 0.07662$; $m3$: $0.26589 \times 0.03423 = 0.00910$.
Residuals (model−data): $m0$: +0.00000, $m1$: −0.00061 (-0.04σ), $m2$: +0.00324 ($+0.20\sigma$), $m3$: −0.02290 (-1.46σ).
Phases: natural dipole peak 194.96° ⇒ $\phi_1 = 55.70 - 194.96 = -139.25°$ (−2.430 rad), Chris's φ: −104.96° → 34.30°; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.858**; no transforms: −0.106; (octupole also given its own phase: 0.862). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.542/\alpha$.

Second branch ($b_0 = 0.00°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{202.902}{1204.27} = 0.16849$, $N = 0.99650 + 0.16849 \times 0.22950 = 1.03517$, $K = P/N = 0.16276$.
Plug back: $m0$: $1.03517(1 - 0.03735) = 0.99650$; $m1$: $0.16849 \times 0.23977 = 0.04040$; $m2$: $0.16849 \times 0.49052 = 0.08265$; $m3$: $0.16849 \times 0.07602 = 0.01281$.
Residuals (model−data): $m0$: −0.00000, $m1$: −0.01213 (-0.78σ), $m2$: +0.00927 ($+0.58\sigma$), $m3$: −0.01919 (-1.22σ).
Phases: natural dipole peak 14.96° ⇒ $\phi_1 = 55.70 - 14.96 = +40.75°$ (+0.711 rad), Chris's φ: 75.04° → 34.30°; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.848**; no transforms: 0.747; (octupole also given its own phase: 0.855). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.326/\alpha$.

B. Numeric: fitted parameters $\kappa = 0.4131$, $D/L = 5.43$, $f_u = 0.0061$, $b_0 = 30.00°$ (bound), $\ell_0 = 14.96°$ (fixed), $\phi_1 = +39.99°$, $\phi_2 = -0.17°$.

Set equal (harmonics of the fitted numeric curve vs data)

m	data $\lvert A_m\rvert$	numeric $\lvert M_m\rvert$	resid.	resid./σ_m	natural peak	transformed peak (data)
0	0.9965	1.0000	+0.0035	+0.30	–	–
1	0.0525	0.0511	−0.0014	−0.09	14.96°	54.95° (55.70°)
2	0.0734	0.0775	+0.0042	+0.26	14.96°	14.78° (14.96°)
3	0.0320	0.0084	−0.0236	−1.50	74.96°	74.96° (66.96°)†

†octupole untouched. (The numeric curve is mean-normalised, so its m=0 is 1 by construction; R^2 uses the curve as is, no rescaling.) Octupole mismatch $\delta_3 = -8.00°$. Chris's φ: 75.04° → 35.05° ($\phi_1 = +0.698$ rad); $\varphi_2 = 60.43°$.
R^2 (71 pts): transformed **0.853**; same parameters, no transforms 0.726; stored no-transform best fit 0.743. $M(< D)/M_{\rm crit} = 2\kappa = 0.826$, $L = 7.90$ Gpc.
Best solution on the other dipole branch (natural dipole flipped to ℓ_0+180°): $R^2 = 0.821$ (no transforms 0.232), $\kappa = 0.132$, $D/L = 300.00$, $f_u = 0.006$, $b_0 = -0.0°$, $\phi_1 = -139.8°$, $\phi_2 = -0.12°$.

$D/r = 4$ ($\varepsilon = 0.2500$, $D = 57.2$ Gpc $= 186.6$ Gly)

A. Analytic: set equal (numbers substituted, $\varepsilon = 0.2500$)

$$0.99650 = N(1 + Kq_0), \quad 0.05253 = NK \cdot 0.2500\,|o_1|, \quad 0.07338 = NK\,|q_2|, \quad 0.03200 = NK \cdot 0.2500\,|o_3|$$

Solve. $\chi^2(b_0)$ has 2 local minima on $[0°, 90°)$:

branch	b_0	q_0	q_2	o_1	o_3	$c_1,\ c_2,\ c_3$	$P = NK$	χ^2	dipole
primary	43.32°	+0.0946	0.2596	−0.7084	−0.0878	0.1771, 0.2596, 0.0220	0.29302	2.684	flipped
second	0.00°	−0.2295	0.4905	+0.7193	−0.2281	0.1798, 0.4905, 0.0570	0.17190	4.388	on axis

Primary ($b_0 = 43.32°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{115.823}{395.28} = 0.29302$, $N = 0.99650 - 0.29302 \times 0.09461 = 0.96878$, $K = P/N = 0.30246$.
Plug back: $m0$: $0.96878(1 + 0.02861) = 0.99650$; $m1$: $0.29302 \times 0.17710 = 0.05189$; $m2$: $0.29302 \times 0.25961 = 0.07607$; $m3$: $0.29302 \times 0.02195 = 0.00643$.
Residuals (model−data): $m0$: +0.00000, $m1$: −0.00064 (-0.04σ), $m2$: +0.00269 ($+0.17\sigma$), $m3$: −0.02557 (-1.63σ).
Phases: natural dipole peak 194.96° ⇒ $\phi_1 = 55.70 - 194.96 = -139.25°$ (−2.430 rad), Chris's φ: −104.96° → 34.30°; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.846**; no transforms: −0.117; (octupole also given its own phase: 0.850). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.605/\alpha$.

Second branch ($b_0 = 0.00°$): $P = \dfrac{\sum w_m A_m c_m}{\sum w_m c_m^2} = \dfrac{187.522}{1090.88} = 0.17190$, $N = 0.99650 + 0.17190 \times 0.22950 = 1.03595$, $K = P/N = 0.16593$.
Plug back: $m0$: $1.03595(1 - 0.03808) = 0.99650$; $m1$: $0.17190 \times 0.17983 = 0.03091$; $m2$: $0.17190 \times 0.49052 = 0.08432$; $m3$: $0.17190 \times 0.05701 = 0.00980$.
Residuals (model−data): $m0$: +0.00000, $m1$: −0.02162 (-1.38σ), $m2$: +0.01094 ($+0.69\sigma$), $m3$: −0.02220 (-1.41σ).
Phases: natural dipole peak 14.96° ⇒ $\phi_1 = 55.70 - 14.96 = +40.75°$ (+0.711 rad), Chris's φ: 75.04° → 34.30°; $\phi_2 = 0°$, $\varphi_2 = 60.08°$; $\delta_3 = -8.00°$ (-24.0° in 3ℓ).
R^2 (71 pts): rotation $+2\ell$: **0.801**; no transforms: 0.724; (octupole also given its own phase: 0.807). $M(< D)/M_{\rm crit} = 2K/\alpha = 0.332/\alpha$.

B. Numeric: fitted parameters $\kappa = 0.6488$, $D/L = 3.13$, $f_u = 3.8e-10$, $b_0 = -30.00°$ (bound), $\ell_0 = 14.96°$ (fixed), $\phi_1 = +39.94°$, $\phi_2 = -0.19°$.

Set equal (harmonics of the fitted numeric curve vs data)

m	data $\lvert A_m\rvert$	numeric $\lvert M_m\rvert$	resid.	resid./σ_m	natural peak	transformed peak (data)
0	0.9965	1.0000	+0.0035	+0.30	–	–
1	0.0525	0.0507	−0.0018	−0.12	14.96°	54.90° (55.70°)
2	0.0734	0.0772	+0.0038	+0.24	14.96°	14.77° (14.96°)
3	0.0320	0.0049	−0.0271	−1.73	74.96°	74.96° (66.96°)†

†octupole untouched. (The numeric curve is mean-normalised, so its m=0 is 1 by construction; R^2 uses the curve as is, no rescaling.) Octupole mismatch $\delta_3 = -8.00°$. Chris's φ: 75.04° → 35.10° ($\phi_1 = +0.697$ rad); $\varphi_2 = 60.45°$.
R^2 (71 pts): transformed **0.837**; same parameters, no transforms 0.711; stored no-transform best fit 0.729. $M(< D)/M_{\rm crit} = 2\kappa = 1.298$, $L = 18.30$ Gpc.
Best solution on the other dipole branch (natural dipole flipped to ℓ_0+180°): $R^2 = 0.767$ (no transforms 0.329), $\kappa = 0.138$, $D/L = 300.00$, $f_u = 0.693$, $b_0 = -0.0°$, $\phi_1 = -139.7°$, $\phi_2 = -0.11°$.

Summary in Chris's variables ($r = 14.3$ Gpc, $\ell_0 = 14.96°$ for all rows; analytic $M/M_{\rm crit}$ uses $\alpha = 1$)

D/r	model	D [Gpc]	D [Gly]	$M/M_{\rm crit}$	D/L	b_0	ϕ_1 [deg]	ϕ_1 [rad]	ϕ_2 [deg]	$R^2_{\rm tf}$	$R^2_{\rm no\,tf}$
2	numeric	28.6	93.3	0.216	300*	0.0°	−139.78	−2.440	−0.12	**0.885**	0.027
	(alt. branch)			0.529	15.24	30.0°	+40.11	+0.700	−0.15	0.879	0.750
	analytic			0.306	–	6.3°	+40.75	+0.711	+0.00	**0.884**	0.752
	analytic (2nd)			0.486	–	36.2°	−139.25	−2.430	+0.00	0.876	−0.088
3	numeric	42.9	139.9	0.826	5.43	30.0°*	+39.99	+0.698	−0.17	**0.853**	0.726
	(alt. branch)			0.265	300*	−0.0°	−139.77	−2.439	−0.12	0.821	0.232
	analytic			0.542	–	40.0°	−139.25	−2.430	+0.00	**0.858**	−0.106
	analytic (2nd)			0.326	–	0.0°	+40.75	+0.711	+0.00	0.848	0.747
4	numeric	57.2	186.6	1.298	3.13	−30.0°*	+39.94	+0.697	−0.19	**0.837**	0.711
	(alt. branch)			0.277	300*	−0.0°	−139.71	−2.438	−0.11	0.767	0.329
	analytic			0.605	–	43.3°	−139.25	−2.430	+0.00	**0.846**	−0.117
	analytic (2nd)			0.332	–	0.0°	+40.75	+0.711	+0.00	0.801	0.724

* at fit bound ($|b_0| \le 30°$, $D/L \le 300$; $D/L = 300$ is effectively a point mass at the CoE). Octupole mismatch $\delta_3 = -8.0°$ in every row (octupole at natural phase about ℓ_0). Numeric rows: $f_u = 0.0345$ ($D/r = 2$), 0.00606 ($D/r = 3$), 3.76e-10 ($D/r = 4$). Analytic rows: N, K = 1.0314, 0.1529 ($D/r = 2$); 0.9820, 0.2708 ($D/r = 3$); 0.9688, 0.3025 ($D/r = 4$) (primary). Data Fourier fit: $R^2 = 0.915$ (ceiling for any $m \le 3$ curve).

Harmonic amplitudes at a glance (data: $|A_1|, |A_2|, |A_3| = 0.0525, 0.0734, 0.0320$)

D/r	num. $\lvert M_1\rvert$	num. $\lvert M_2\rvert$	num. $\lvert M_3\rvert$	an. $\lvert M_1\rvert$	an. $\lvert M_2\rvert$	an. $\lvert M_3\rvert$
2	0.0462	0.0798	0.0207	0.0530	0.0764	0.0177
3	0.0511	0.0775	0.0084	0.0519	0.0766	0.0091
4	0.0507	0.0772	0.0049	0.0519	0.0761	0.0064

Reference: numeric with ℓ_0 also free (a third free phase in effect): $D/r = 2$: $R^2 = 0.894$ ($\ell_0 = 130.1°$, $\delta_3 = -3.1°$); $D/r = 3$: $R^2 = 0.858$ ($\ell_0 = 6.9°$, $\delta_3 = +0.1°$); $D/r = 4$: $R^2 = 0.839$ ($\ell_0 = 187.0°$, $\delta_3 = -0.0°$). At $D/r = 2$ and 4 the free ℓ_0 runs far from the data axis (130°, 187°) and the transforms then re-point $m1$ and $m2$. The gain over fixed ℓ_0 is ≤ 0.01 in R^2.

Observations. (1) With the two phases free, both primary solutions match $|m1|$ and $|m2|$ to within 0.3σ at every D/r; the one exception is the numeric fit at $D/r = 2$ ($\mp 0.4\sigma$). Where they fail, both fail in the octupole. (2) The octupole is the failure: $|m3| \propto \varepsilon$ in the analytic form, and the numeric one is similar ($0.021, 0.008, 0.005$ vs data 0.032), and its natural phase is 8° off. This is why $R^2_{\rm tf}$ falls from $D/r = 2$ to 4. (3) The transformed R^2 values are all 0.84–0.89, below the $m \le 3$ Fourier ceiling of 0.915. In both methods the rotation is $\phi_1 \approx +40°$ (or $-140°$ on the flipped branch) and $\phi_2 \approx 0°$. (4) Without transforms, the on-axis branches keep $R^2 \approx 0.71$–0.75 and the flipped branches collapse ($R^2 \le 0.33$). For $D/r = 3, 4$ the analytic weighted-LSQ optimum is the flipped branch ($b_0 \approx 40°$) by a small χ^2 margin.

Caveats. The rotation ϕ_1 and the 2ℓ phase ϕ_2 (and ℓ_0) are free parameters; they are not predicted by the axisymmetric physics. With them free, only amplitudes are being tested, and $|m1|, |m2|$ are matched almost automatically. The 71 points come from overlapping 90° sectors stepped by 5°, so there are only ∼16 independent values: R^2 overstates the evidence, and differences of a few $\times 0.01$ are not significant. The analytic form is a point-mass tidal series truncated at $O(\varepsilon)$. It misses the dipole contributed by the extended fitted density in the numeric model, so analytic K and numeric κ are not the same quantity ($\alpha \ne 1$ in general). Several numeric parameters sit on their bounds ($|b_0| = 30°$, or $D/L = 300$), and the data sector weighting assumes clusters uniform on the sky outside $|b| \le 20°$.